

# Vinod Kumar Gorle

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## **SUMMARY**

AI / GenAI Engineer with nearly 4 years of experience building AI applications using LLMs, VLMs, RAG, multimodal AI, semantic search, and intelligent retrieval. Experienced with LangChain, Azure AI Foundry, Azure AI Search, Azure Machine Learning, FastAPI, LLaMA-3, Mistral, Qwen-VL, LLaVA, BGE-M3, SigLIP, FAISS, and Sentence Transformers. Skilled in AI architecture, query understanding, hybrid retrieval, embeddings, VLM-based image analysis, prompt engineering, model evaluation, retrieval optimization, secure deployment, and enterprise AI solutions.

## **SKILLS**

### **GenAI, LLMs & RAG:**

Generative AI, Agentic AI, LLMs, VLMs, RAG, Multimodal AI, AI Architecture, LangChain, LangGraph, Prompt Engineering, Prompt Injecting, Query Understanding, Intent Classification, RAG Evaluation

### **Search & Retrieval:**

Semantic Search, Lexical Search, Hybrid Search, Vector Search, Multimodal Retrieval, Embeddings, Reranking, Query Expansion, Retrieval Optimization, FAISS, Azure AI Search, BGE-M3, SigLIP, all-MiniLM-L6-v2, Sentence Transformers, Text-to-SQL, Vector Database, Voice-To-Text

### **AI/ML & Computer Vision:**

Machine Learning, Deep Learning, PyTorch, Scikit-learn, NLP, Computer Vision, YOLOv5, OpenCV, OpenVINO, Object Detection, Object Tracking, Video Analytics, Edge AI, Model Optimization, FP16, TensorFlow, Hugging Face, PyTorch, Data pre-processing, Feature Engineering, Advanced Data Analysis.

### **Azure AI & Cloud:**

Azure AI Foundry, Azure Machine Learning, Azure AI Search, Azure Blob Storage, Microsoft Entra ID, Azure OpenAI, Gemini models, Managed Endpoints, AI Model Deployment, Model Lifecycle Management

### **Cloud & DevOps:**

Microsoft Azure, Azure App Services, Azure Functions, Azure Storage, Azure Compute, Cloud AI Architecture, Cloud Deployment, Docker, Linux, Git, GitHub, CI/CD, API Deployment, AI/ML Model Serving, Scalable AI Inference

### **Backend & Data:**

Python, FastAPI, Flask, REST APIs, API Architecture, Asynchronous Processing, MS-SQL, MySQL, Database Design, Query Optimization, Data Pipelines, Dagster Automation

### **AI Development:**

Cursor, OpenAI Codex, GitHub Copilot, AI Coding Agents, AI-Assisted Development, Code Generation, Code Refactoring

### **Full-Stack:**

React JS, Node JS, JavaScript, HTML, CSS, jQuery, Bootstrap, WordPress

## **EXPERIENCE**

## **SOFTWARE ENGINEER — Xinthe Technologies (June 2023 – Present)**

### **❖ Enterprise AI Logo Search & Multimodal Retrieval Platform**

**Azure AI Foundry | Azure Machine Learning | Azure AI Search | LangChain | LLMs | VLMs | SigLIP | BGE-M3  
| FAISS | SQL Server | Python | FastAPI | Azure Blob Storage | Microsoft Entra ID**

- Architected an enterprise multimodal AI search platform using LLMs, VLMs, RAG, BGE-M3, SigLIP, semantic/vector search, hybrid retrieval, metadata filtering, and image similarity.
- Built end-to-end ingestion, VLM metadata generation, embedding/indexing, LLM query understanding, lexical/semantic/visual retrieval, query expansion, ranking and reranking pipelines using LangChain, FAISS, SQL Server and FastAPI.
- Evaluated LLM/VLM models for query-to-SQL, intent classification, captioning and semantic retrieval, achieving 85–90% accuracy; deployed with Azure AI Foundry, Azure ML, Azure AI Search, Blob Storage and Entra ID.

### **❖ Enterprise Document Q&A & RAG Platform**

**FastAPI | LLaMA-3 | Mistral | Ollama | LangChain | RAG | all-MiniLM-L6-v2 | Sentence Transformers | FAISS | Python | SQL**

- Developed a self-hosted GenAI Document Q&A platform using LLaMA-3, Mistral, Ollama, LangChain, FastAPI and Python, supporting offline LLM inference, RAG and knowledge-grounded question answering.
- Designed an end-to-end RAG pipeline covering document ingestion, parsing, preprocessing, chunking, embeddings, FAISS vector indexing, Top-K semantic retrieval, query understanding, context construction, prompt engineering, and LLM response generation.
- Optimized retrieval and response quality using all-MiniLM-L6-v2, similarity thresholds, chunk size/overlap, Top-K/Top-P, reranking and retrieval parameters; evaluated groundedness, faithfulness, relevance, correctness, accuracy, hallucination, latency and inference performance.

### **❖ Python Backend & AI/ML Application Architecture**

**Python | FastAPI | Flask | Django | PyTorch | Scikit-learn | Machine Learning | Deep Learning | NLP | Computer Vision | Transformers | Model Training | Model Inference | Model Evaluation | REST APIs | MS-SQL | Async Processing**

- Designed scalable Python backend and AI/ML architectures using FastAPI, Flask, Django, PyTorch, and Scikit-learn for data processing, model training, evaluation, inference, NLP, Computer Vision, and LLM/VLM services.
- Built production-ready enterprise services with MS-SQL, REST APIs, asynchronous workflows, authentication, model deployment, CI/CD, vector search, embeddings, and GenAI integrations, focusing on scalability, performance, and reliability.

### **❖ Computer Vision & Edge AI Architecture**

**YOLOv5 | OpenCV | OpenVINO | Object Detection | Tracking | Edge AI**

- Designed real-time computer vision and video analytics pipelines for object detection, people counting, surveillance, tracking, ROI-based analysis, and event detection, achieving 90%+ counting accuracy through optimized tracking and post-processing.
- Architected edge AI inference using Intel OpenVINO with FP16/INT8 optimization for low-latency, high-FPS deployment, integrating vision inference with backend APIs and real-time analytics while optimizing latency, throughput, and hardware utilization.

## ❖ **Full-Stack & Web Application Engineering**

**React | JavaScript | PHP | WordPress | HTML | CSS | jQuery**

- Designed modular full-stack architectures integrating React/JavaScript frontends with Python backend services, REST APIs, enterprise data workflows, and third-party integrations.
- Developed reusable React/JavaScript components and responsive interfaces using jQuery, HTML, CSS, and Bootstrap for interactive enterprise dashboards and business applications.
- Built and customized WordPress applications with reusable UI components and API integrations, working across frontend, backend, API, and database layers for end-to-end application development.

## **SOFTWARE ENGINEER INTERN — Xinthe Technologies (Nov 2022 – May 2023)**

- Performed QA testing, model validation, inference benchmarking, and performance evaluation for Automatic License Plate Recognition (ALPR) systems using live and recorded video streams, identifying detection and inference issues to improve model reliability.
- Designed and developed real-time people counting and video analytics pipelines using Intel OpenVINO, OpenCV, and computer vision techniques for surveillance and edge AI applications.
- Created and maintained annotated datasets, performed model validation, error analysis, inference output verification, and performance evaluation to improve computer vision model accuracy and deployment readiness.

## **CERTIFICATIONS & TRAINING**

- Develop Generative AI Solutions with Azure OpenAI Service – Microsoft Learn
- Develop Generative AI Apps in Azure – Microsoft Learn
- Deploy and Manage Machine Learning Models with Azure Machine Learning – Microsoft Learn
- Explore Azure AI Foundry – Microsoft Learn
- Python Programming – Cisco
- C Language – Cisco
- Dagster Essentials Certification – Dagster University
- Ultimate MS-SQL Bootcamp & Complete JavaScript Course – Udemy

## **WEBSITES & SOCIAL LINKS**

- Portfolio: <https://portfolio-vinodkumargorle.vercel.app/>
- LinkedIn: <https://www.linkedin.com/in/vinodkumargorle8367/>
- GitHub: <https://github.com/vinodkumargorle/>

## **EDUCATION**

- **MCA – Master of Computer Applications (2020–2022)**  
Gayatri Vidya Parishad College of Engineering, Visakhapatnam, GPA: **7.6 / 10**
- **B.Sc. – Computer Science (2017–2020)**  
Sri GCSR College, Rajam, Andhra Pradesh, GPA: **7.8 / 10**